



Data Analytics

Session 3: Introduction to Data Analytics

Prof. Dr. Wolfgang Kratsch

Augsburg Technical University of Applied Sciences,
Faculty of Computer Science,
Professor of Applied AI in Digital Value Creation

FIM Research Center for Information Management

Branch Business & Information Systems Engineering of the Fraunhofer FIT

www.fim-rc.de

www.wirtschaftsinformatik.fraunhofer.de

Lecturer: Prof. Dr. Wolfgang Kratsch

Curriculum Vitae

- Bachelor and Master's in Information Systems, University of Augsburg, graduated in 2017
- PhD in Process Mining, University of Bayreuth, completed in 2020
- Co-founder of THA/Fraunhofer FIT spin-off, credium GmbH, based in Augsburg, responsible for AI & Data Science
- Postdoctoral researcher at THA with Prof. Häckel, from 2021 to 2023, at the TTZ Data Analytics & Fraunhofer FIT, focusing on Data and Process Science
- Since February 2023: Professor for Applied AI at THA / Fraunhofer FIT



Teaching Concept

- Highly interactive teaching with digital elements, utilizing the Inverted Classroom approach
- Enabling individual learning paths with continuous monitoring of learning progress
- Numerous practical and application-oriented examples (e.g., from projects)

Content of today's lecture and tutorial

		02:00 p. m. - 03:30 p. m. (Room: W322)	03:40 p. m. - 05:10 p. m. (Room: W322)
1	19.03.2025	Introduction to Python	Python Setup & Python Basics
2	26.03.2025	Self-Study Session	
3	02.04.2025	Introduction to Data Analytics	Discussion of Self-Study Exercises
4	09.04.2025	Data Engineering & Management	Data Engineering with Python
5	16.04.2025	Modeling I	Modeling with Python I
6	23.04.2025	Modeling II	Modeling with Python II
7	30.04.2025	Introduction to Micro Projects	Evaluation
8	01.05.-20.05.2025	Micro Project Phase	
9	21.05.2025	Micro Project Presentations	
10	28.05.2025	Geospatial Data Analytics (Guest Lecture)	Geospatial Data Analytics
11	11.06.2025	Storytelling with Data (Guest Lecture)	Storytelling with Data
12	25.06.2025	Advanced Use Cases & Frontier Topics	Q&A Session
13	tba	Exam (60 min, written test, no accompanying materials permitted)	

Contents of today's lecture



What is Data Analytics and how to characterize it?



What is Data Analytics and how can we classify AI?



What is CRISP-DM and how do the six steps look like?

What is Data Analytics?

Gaining sensible business insights from analyzing Data

„So what’s getting ubiquitous and cheap? Data. And what is complementary to data? Analysis.“
- Hal Varian (2008)

„Information is the oil of the 21st century, and analytics is the combustion engine“
- Peter Sondergaard (Gartner Group, 2008)



„Organizations with data-driven decision-making have output that is 5-6% higher than without.“
- Erik Brynjolfsson (MIT, 2011)

„I shall not [...] attempt further to define [...] that shorthand description; and perhaps I could never succeed in intelligibly doing so. But I know it when I see it [...]“
- Potter Stewart, US Supreme Court (1964)



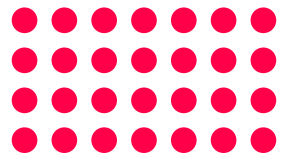
“Data is the information asset characterized by such a high volume, velocity and variety to require specific technology and analytical methods for its transformation into value.” (Mauro et al., 2016)



Data Business Analytics deals with the extraction of economic knowledge from complex data sources and its sensible use for organizational purposes.

How to characterize Data - The 5 V's

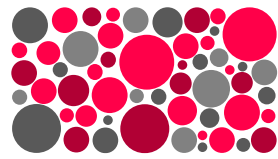
„Volume“



„Data at Rest“

The vast amount of data generated every second, which results in data sets being too large to be stored and analyzed by using traditional database technologies.

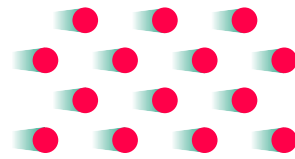
„Variety“



„Data in Many Forms“

The diversity of data. Due to the spread of new technologies, data analytics has to deal with data from various sources being available in different forms, structured and unstructured.

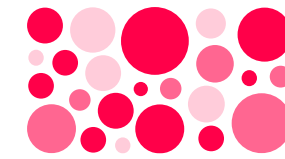
„Velocity“



„Data in Motion“

Data generation speed and pace of data flow. Data is in continuous motion through e.g. business processes and human interaction with social media.

„Veracity“



„Data in Doubt“

Doubts about data cleanliness and credibility. Data diversity makes data control difficult resulting in the data being biased and noisy.

„Value“

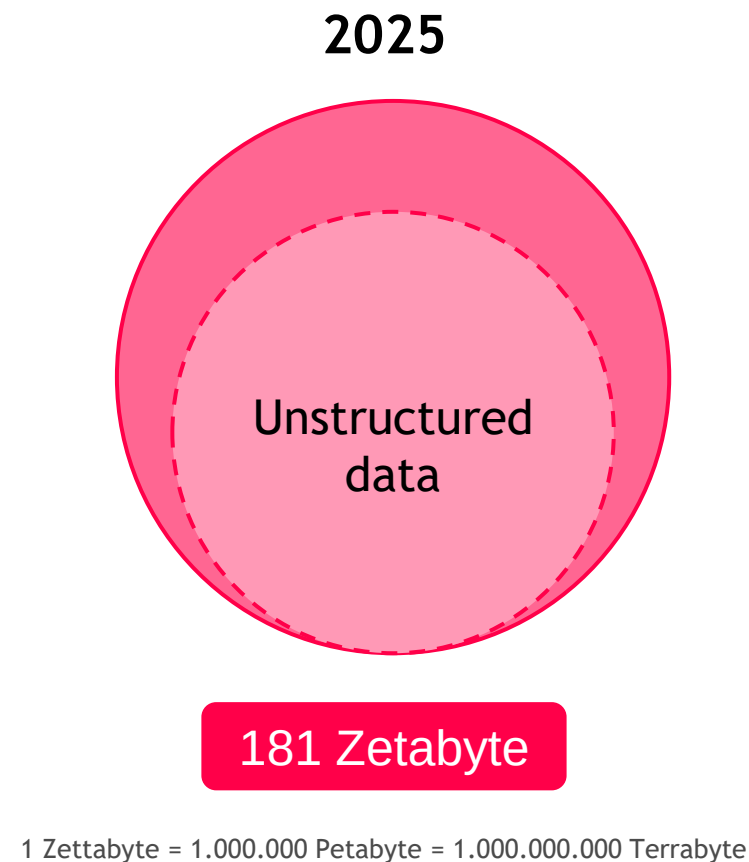
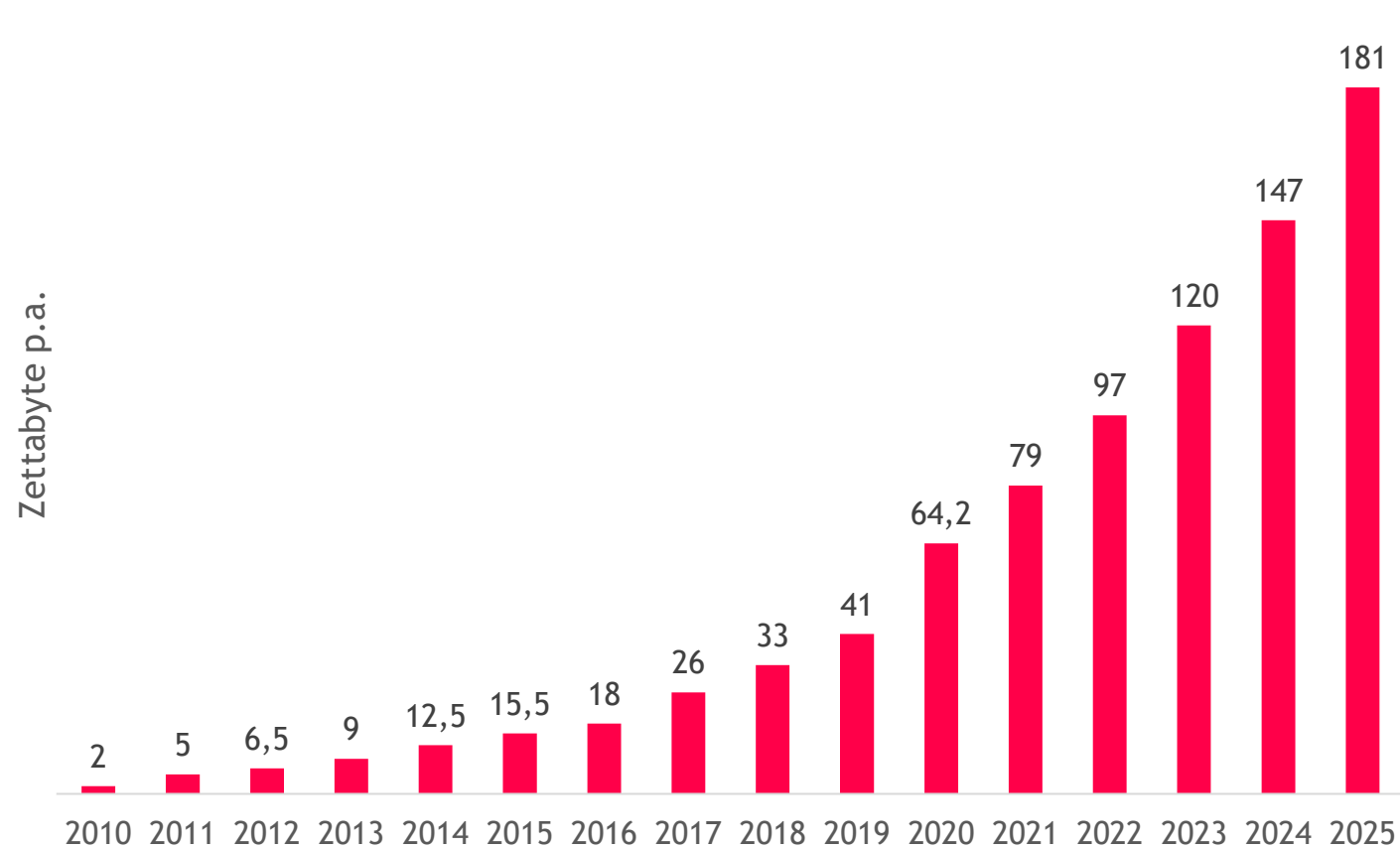


„Data in Limbo“

An organization's ability to translate data into value. Organizations should critically assess the business value of whether collecting and leveraging data brings any value to their business.

[Partially based on Mauro et al. (2016)]

The volume of data generated, consumed, copied and stored is projected to exceed 180 ZB by 2025



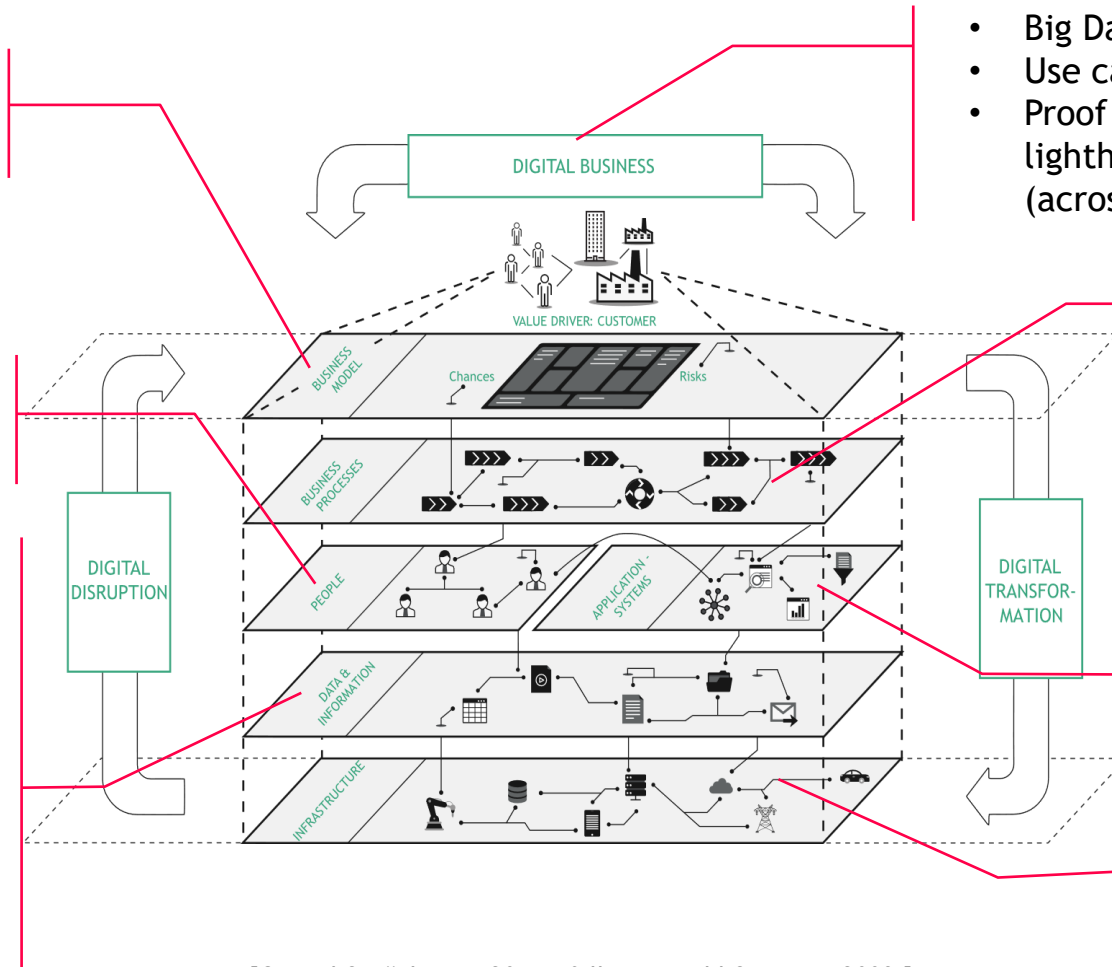
<https://www.cio.com/article/220347/ai-unleashes-the-power-of-unstructured-data.html> | <https://www.statista.com/statistics/871513/worldwide-data-created/>

Today, a value-oriented application of Data Analytics affects all organizational levels

- Data-based Services
- Data monetization
- Risk analysis

- Education and training
- Big-Data-Organization
- Change management

- Data curation (internal and external)
- Data and information life cycle management
- Data integration
- Data quality management
- Data protection (by design)
- Data security



[Gimpel & Röglinger (2017), following Buhl & Kaiser (2008)]

Contents of today's lecture



What is Data Analytics and how to characterize it?



What is Data Analytics and how can we classify AI?



What is CRISP-DM and how do the six steps look like?

The difference between Data Science and Artificial Intelligence



**Mat Velloso**
@matvelloso

Folge ich

Difference between machine learning and AI:

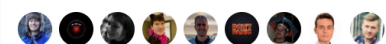
If it is written in Python, it's probably machine learning

If it is written in PowerPoint, it's probably AI

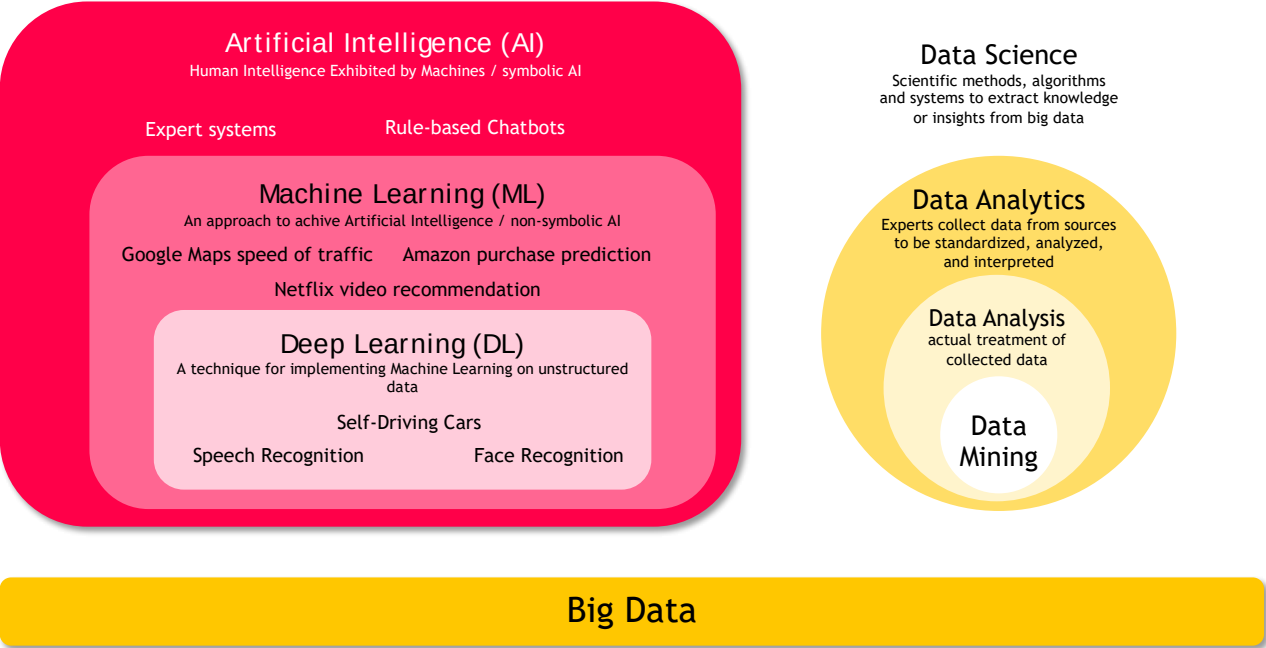
 Tweet übersetzen

17:25 - 22. Nov. 2018

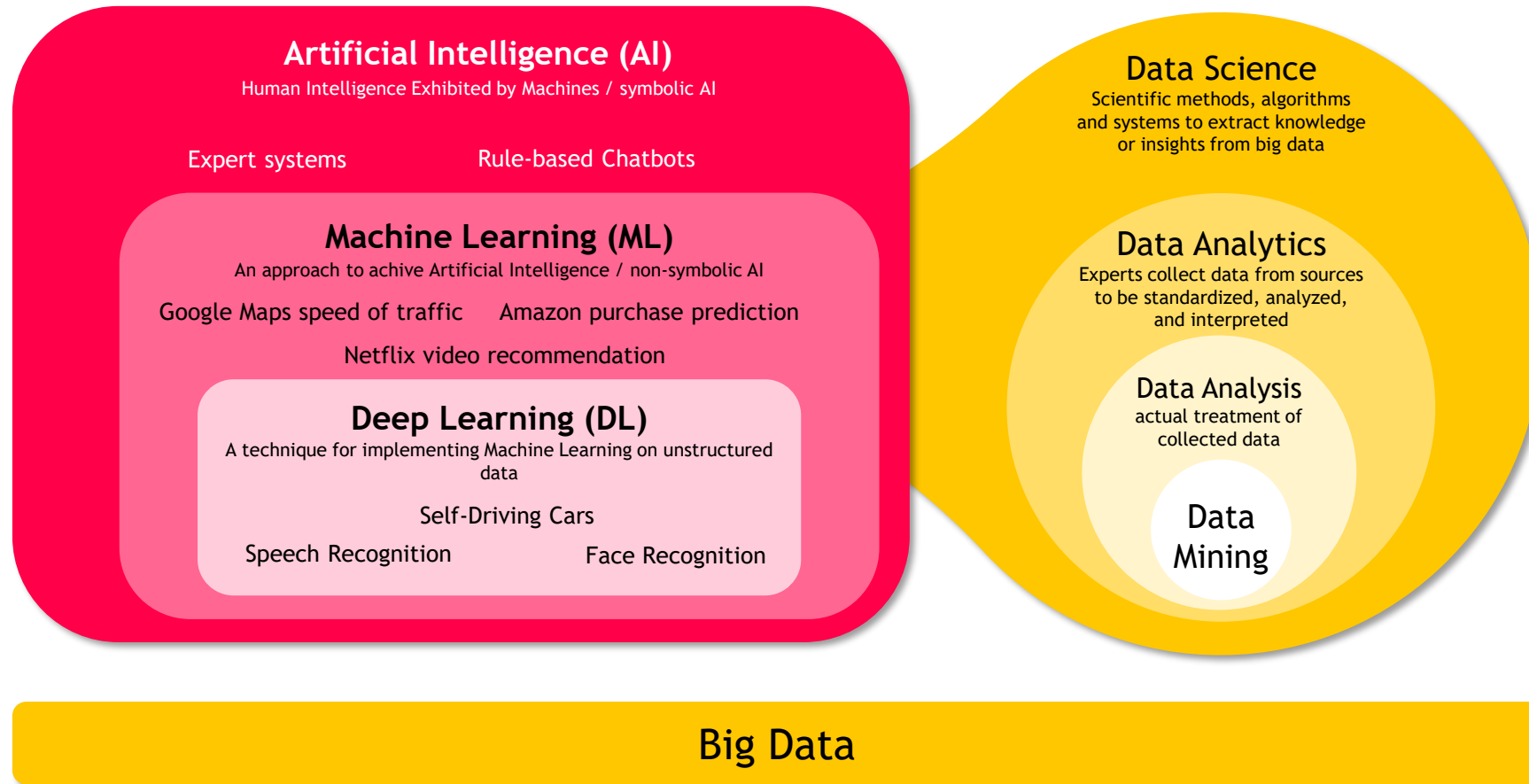
4.054 Retweets 12.485 „Gefällt mir“-Angaben



 120  4,1 Tsd.  12 Tsd. 

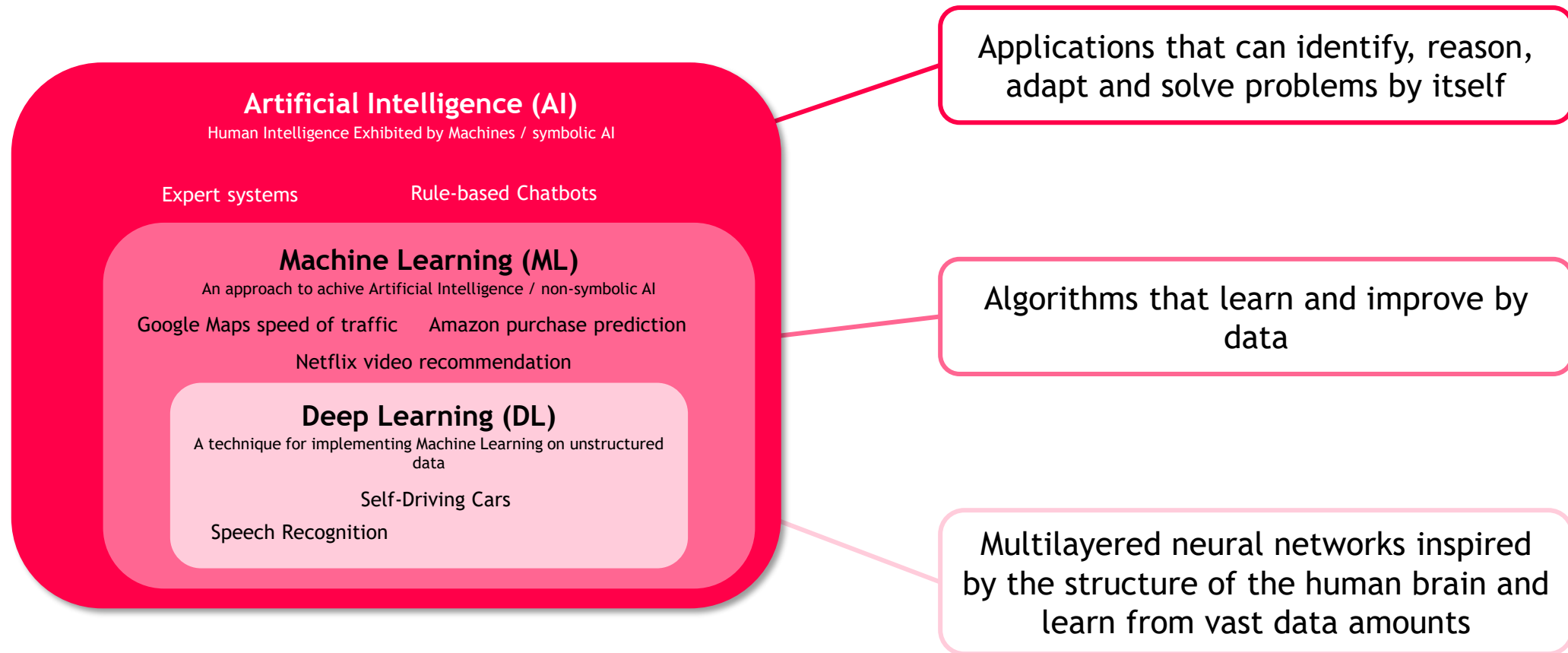


Differentiation of key terms

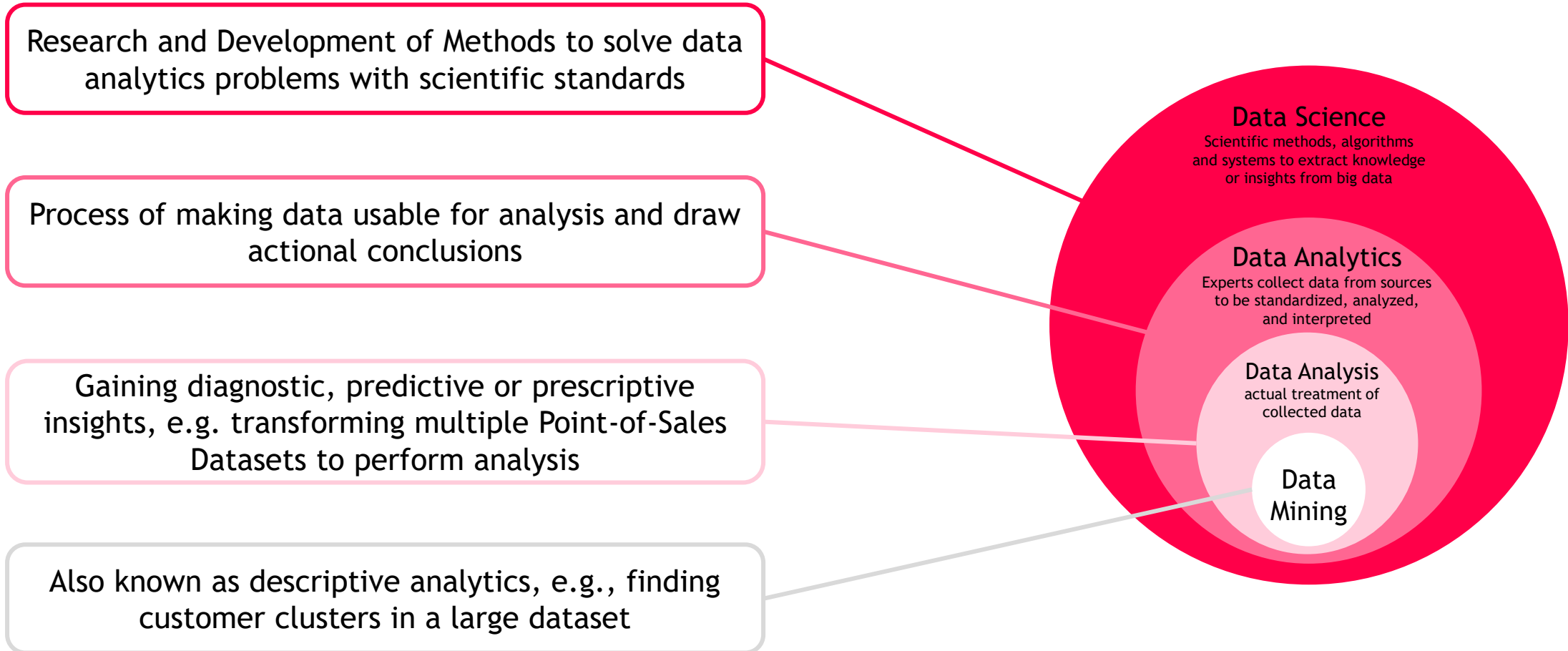


Inspired by <https://www.linkedin.com/pulse/how-make-simple-explain-ai-ml-dl-together-data-science-vollmer/>

Differentiation of key terms - Artificial Intelligence



Differentiation of key terms - Data Science



Four developments allow the application of Data Analytics



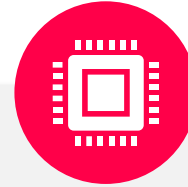
Necessary data is available and usable

Digitization produce data that can be used by Big Data technologies for AI applications.



Machine learning algorithms have been improved

In recent years, significant progress has been made through deep learning algorithms.



Cloud services provide the computing power you need

Cloud services enable a fast, flexible and affordable use of computing resources without major investments. (Scalable and on demand)



The application of Data Analytics is no rocket science

High-performance (open source) toolkits and libraries are available.

The various definitions of AI can be classified in four categories

Thinking Humanly

The idea that AI automates activities which are associated with human thinking (Bellman 1978; Haugeland 1987)

Thinking Rationally

The idea of logic in computational models, i.e. making correct inferences from certain premises

Acting Humanly

The idea that AI automates operations for which people need intelligence (Kurzweil 1990)

Acting Rationally

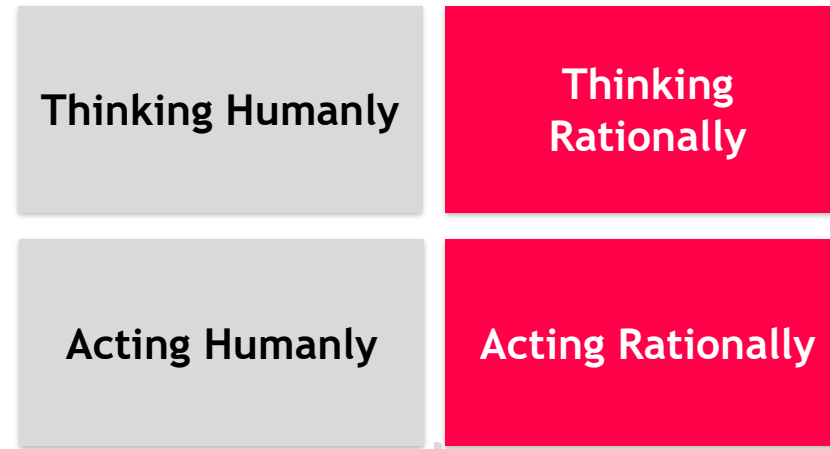
Incorporates uncertainty in situations in which correct inferences are not unambiguously given. In such situations, a rational agent acts upon the goal of achieving the best possible outcome

[Based on Russel and Norvig (2016)]



AI research (and also today's AI applications) is mainly concerned with rationality rather than mimicking human thinking (Sweeney, 2003).

Two propositions for differentiating AI: Strong and weak AI



Strong AI (general purpose)

Concerns the creation of general artificial intelligence. Thereby, development goes so far that the AI's intellectual capabilities and functions mimic the human ones. **While weak AI is often underestimated, the contrary holds true for strong AI.**

Weak AI (problem specific)

Concerns mastering concrete application problems of human thinking by simulating intelligent behavior. Based on the idea that AI is usually able to complete specific tasks quite reliably. AI benefits from being able to learn from large amounts of data and to quickly process information. **However, we should not labor under the misapprehension that the system is actually intelligent.**

Contents of today's lecture



What is Data Analytics and how to characterize it?

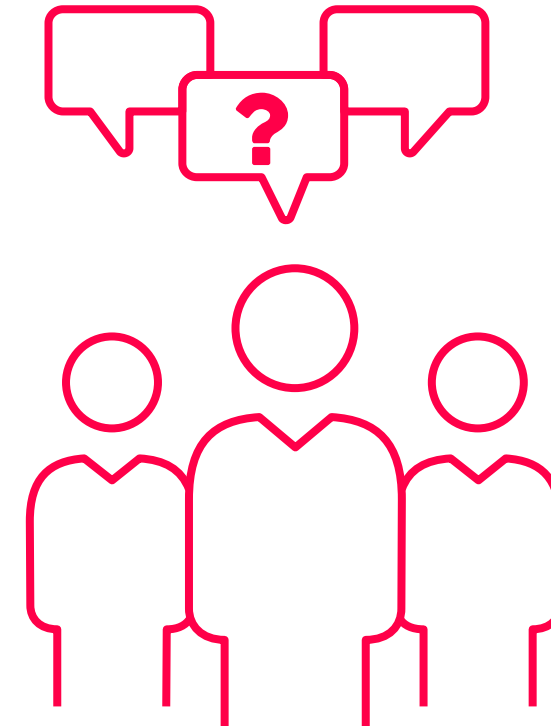


What is Data Analytics and how can we classify AI?

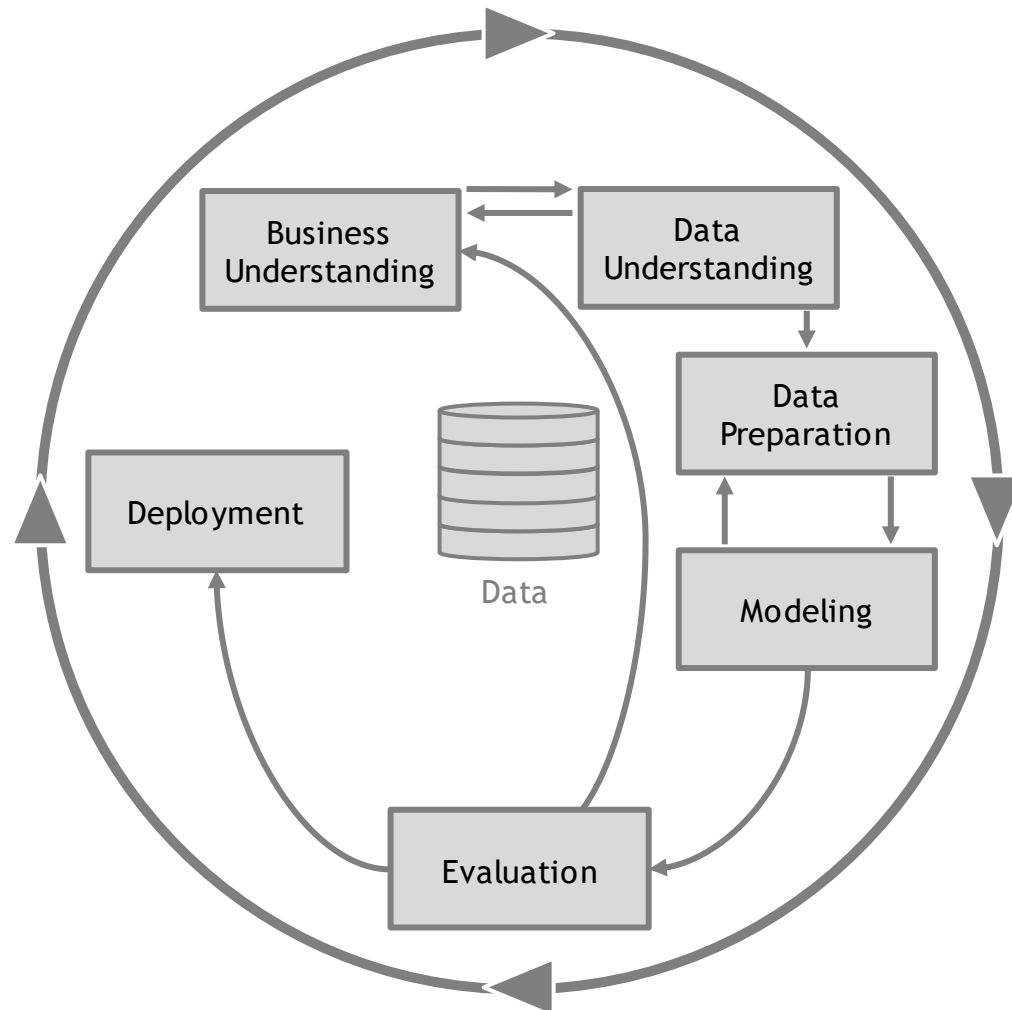


What is CRISP-DM and how do the six steps look like?

What are possible solutions to manage the complexity of Data Analytics?



Data Analytics requires a holistic management process



CRISP-DM **covers the entire analytics process** from target setting to the generation of results

[Based on Chapman et al. (2000)]

What is CRISP-DM and why to use it for Data Analytics?

CRISP-DM

- Cross-Industry Standard Process for Data Mining is the open analytics process standard which is most-widely used
- Originally developed by Data Mining specialists from DaimlerChrysler, SPSS, and NCR and is industry, tool, and application-neutral
- The order of stages is generally not fixed
- Allows for (and recommends) an iterative process considering that answering one business question may lead to new questions

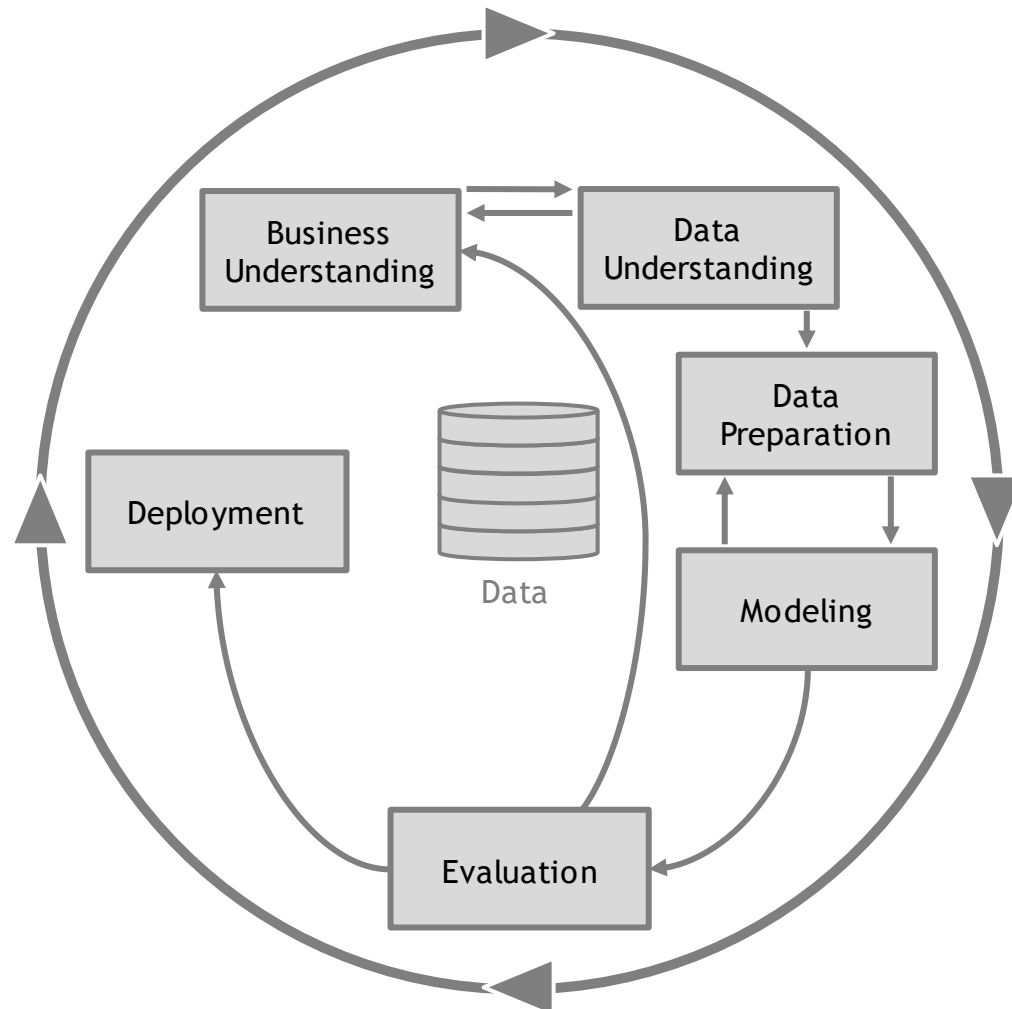
Advantages of a process standard

- not constantly falling back on special processes
- Facilitates understanding
- Improves the quality of results
- Establishes common terms and references
- Improves communication to third parties



CRISP-DM has an open standard and the methodology is available for free (<http://www.crisp-dm.eu>)

In a first step, the data analyst has to understand the business and set the project's goals



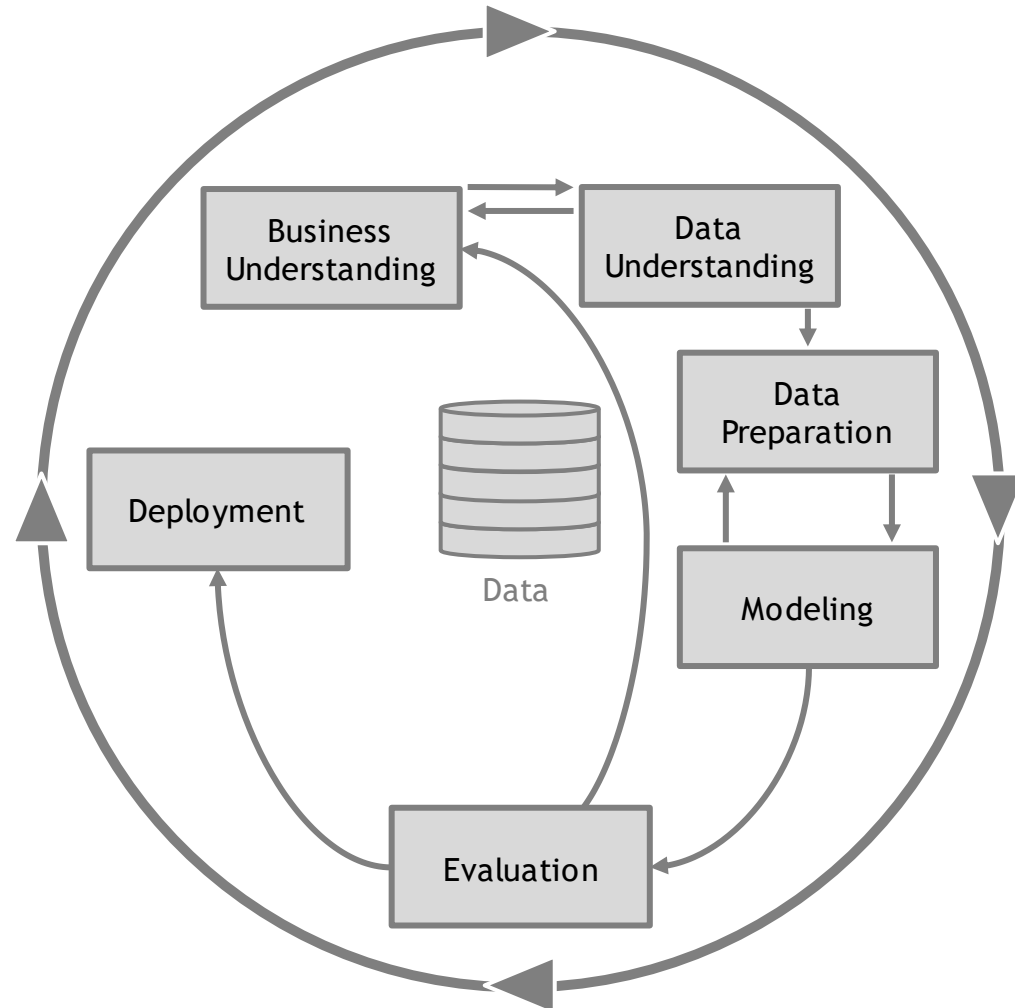
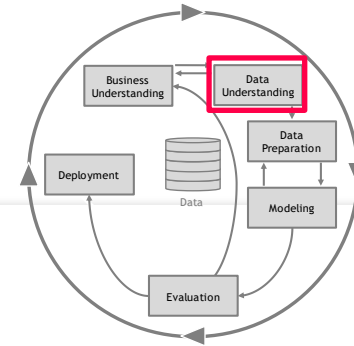
Determination of business objectives

Specification of the data analytics target

Development of a preliminary project plan

[Based on Chapman et al. (2000)]

Secondly, the data analyst should build a basic understanding of data



Selection of relevant
databases

Development of data
understanding

[Based on Chapman et al. (2000)]

Data integration und quality are key success factors in the data analysis process



Challenges

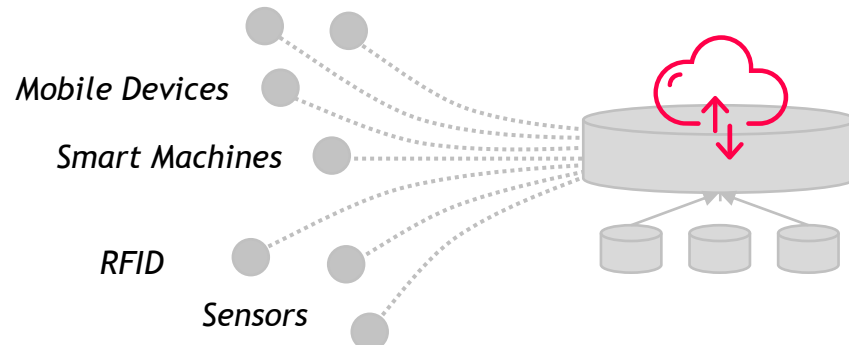
Loads of data have to be well organized & used. Integrated, homogenous data base as competitive advantage.

- Unprecedented volume, variety, velocity and partially questionable veracity of data
- Multitude of external and internal data sources

Possible Approach

- Well organized meta data, social data, mobile data, machine to machine data etc.
- Data warehouse and business intelligence solutions
- Intelligent data quality management processes
- Cleaning of irrelevant data (trash in, trash out)
- Track & trace data
- Access to and integrations of additional suitable external data sources

Towards the data driven enterprise



Fast decisions

Improved Products & Service

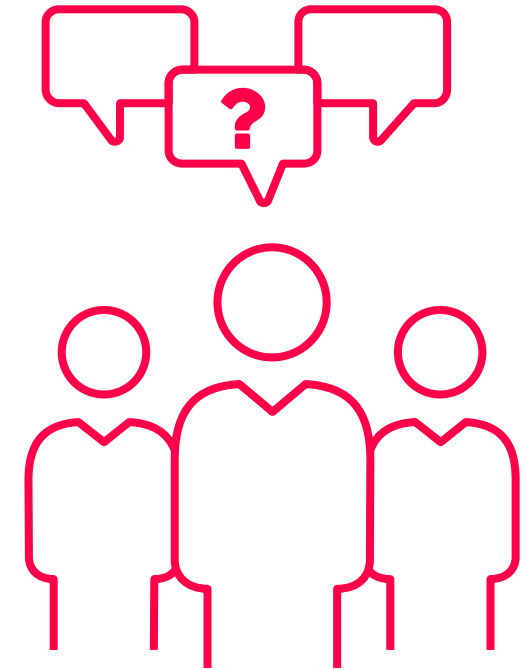
Improved Processes

What is the appropriate degree of data quality?

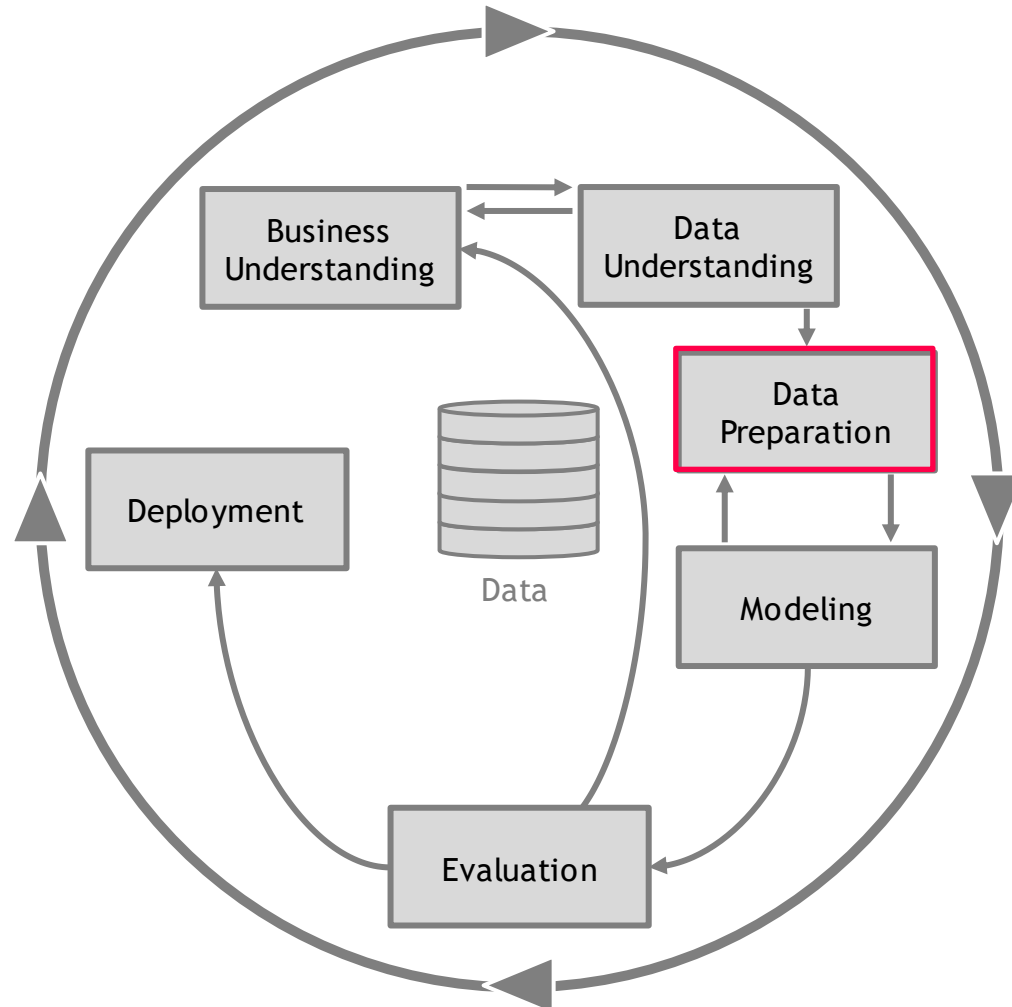
Is there anything like perfect data quality?

What are the factors that have to be considered?

Can you imagine any negative aspects of increasing data quality?



The third step - data preparation - consumes up to 80% of the project time



Data selection

Data cleansing

Data transformation

[Based on Chapman et al. (2000)]

Data preparation determines the efficiency of the analytics model



**Data preparation will be part of your case study task.
Thus, you only find some general hints below:**

Data selection:

- Based on the understanding that has previously been built, only select those attributes that are important for your case study task
- Consider the distribution of values

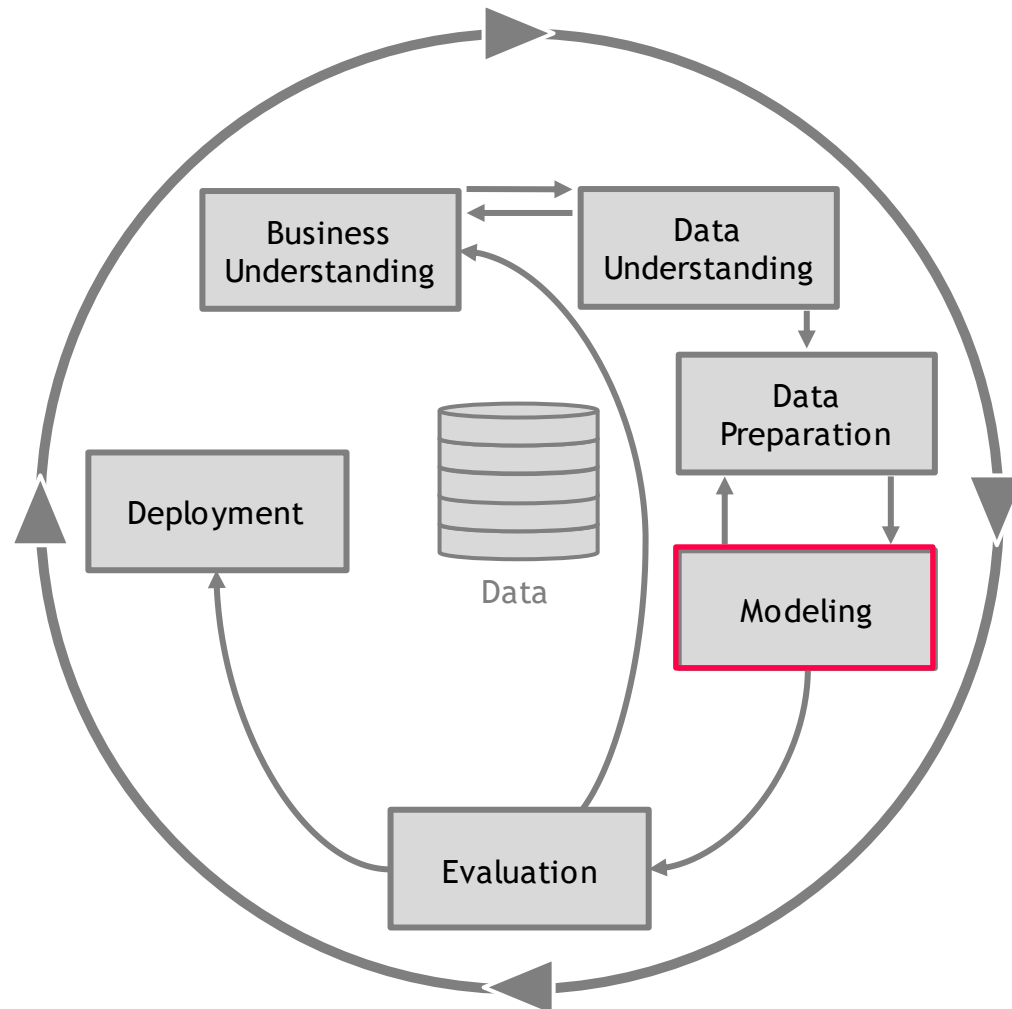
Data cleansing:

- Decide how to handle errors and outliers in the data
- Reducing the data by selecting subsets could be necessary
- If necessary, decide upon an appropriate estimation procedure for missing values

Data transformation:

- You might want to scale, standardize, aggregate and/or recode some attributes (e.g. introducing seasonality or time windows)

Data modeling, as a forth step, is an iterative process that might require repetitive data preparation



Selection of modeling technique

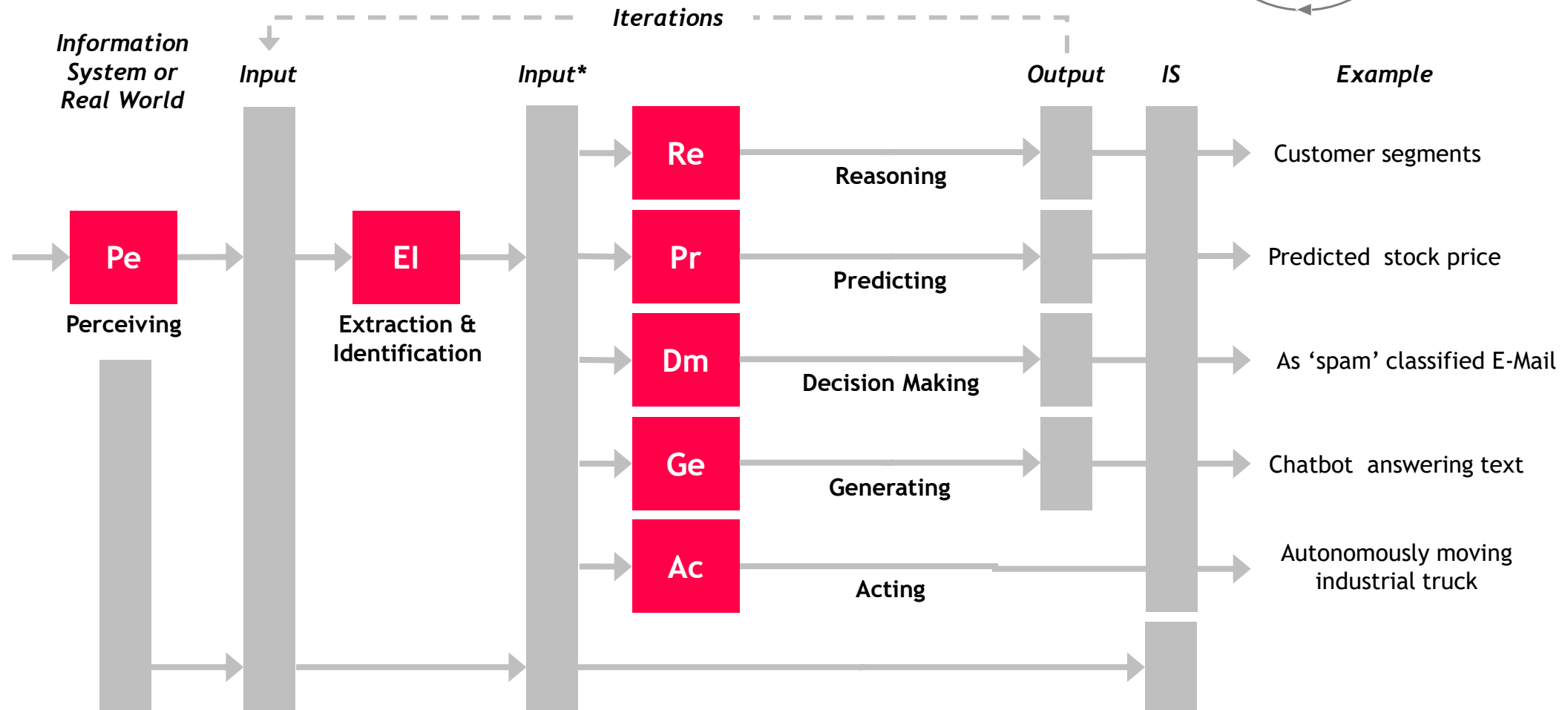
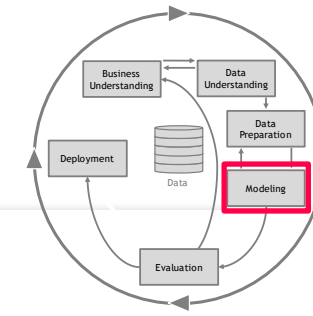
Generation of test design

Model construction

Model evaluation

[Based on Chapman et al. (2000)]

Applied AI - a breakdown of functionality

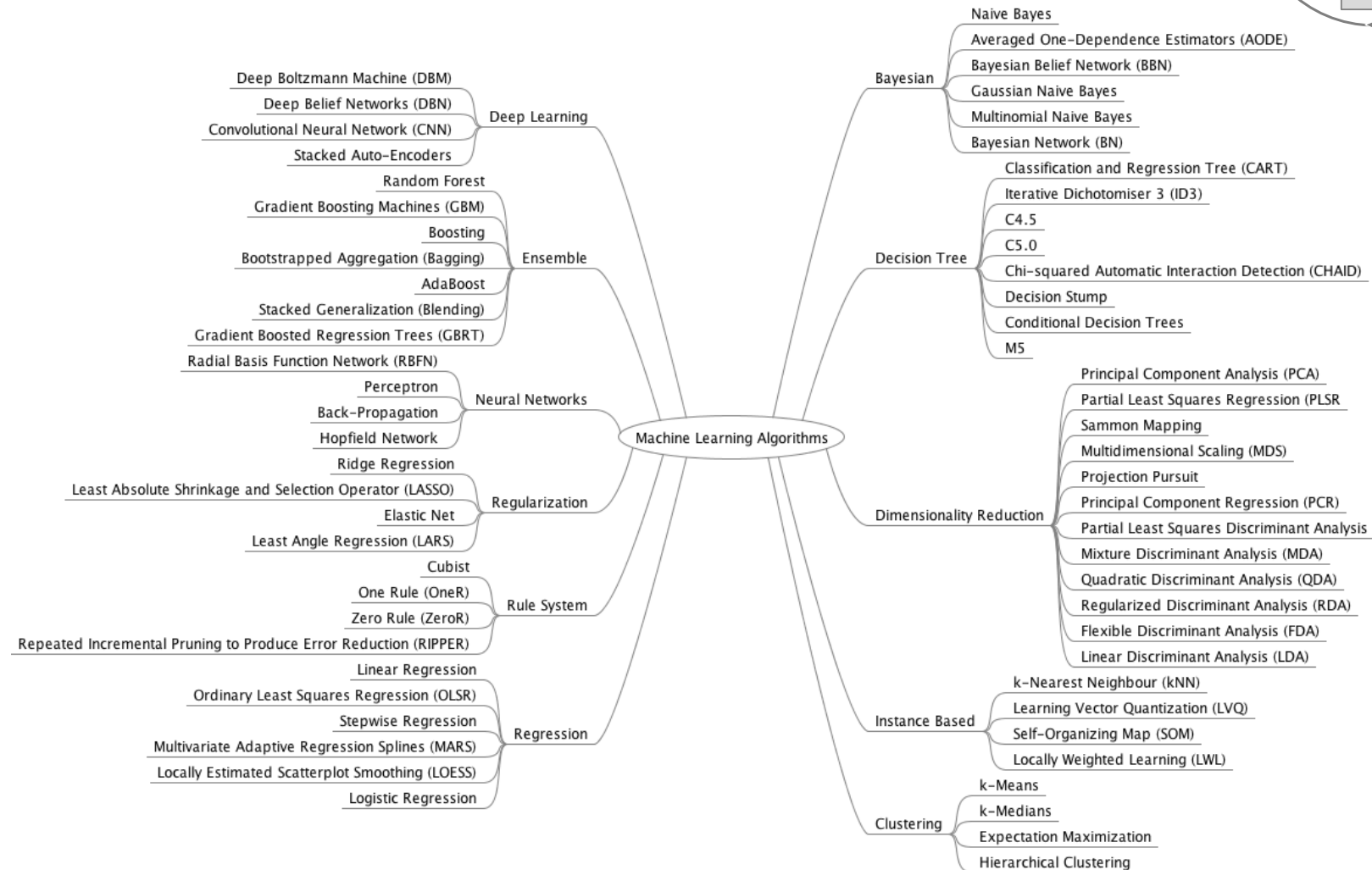


[Hofmann et al. (2020)]

Big-Data-Analytics usually requires a combination of various algorithms

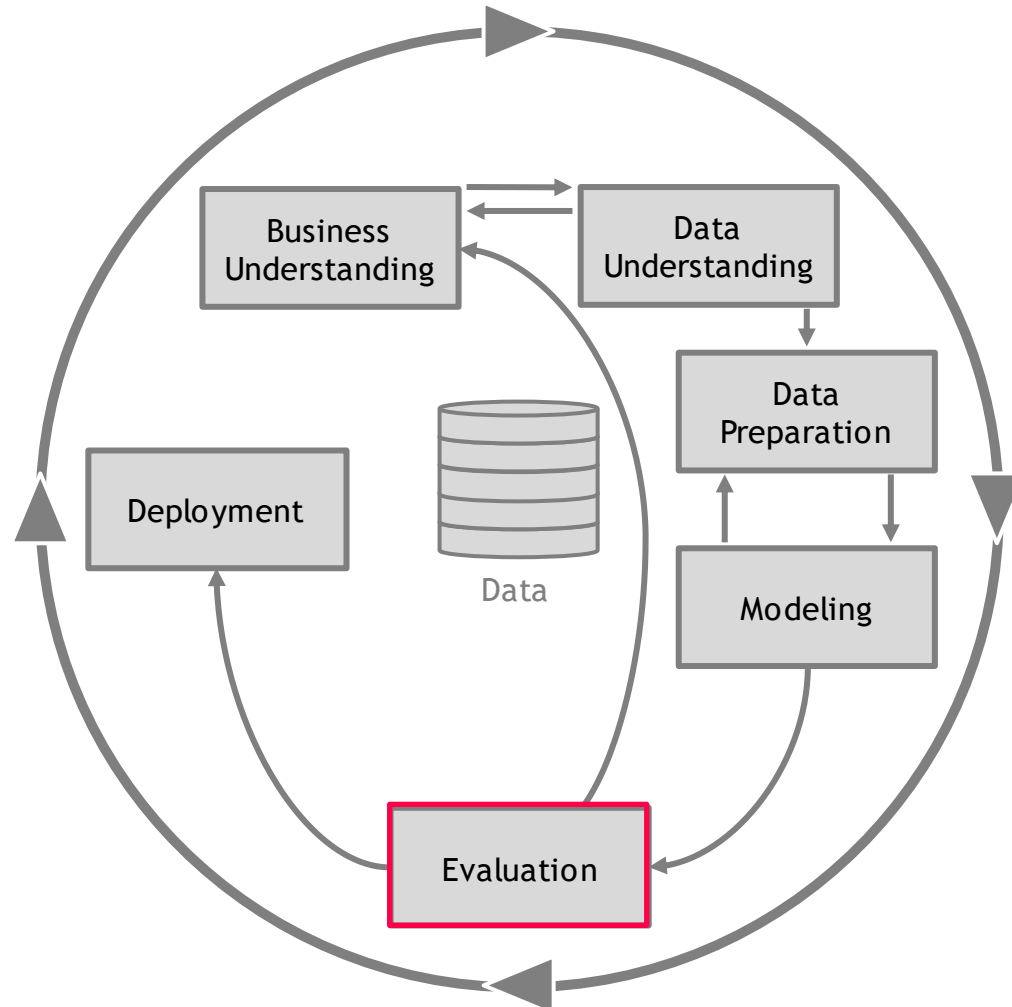
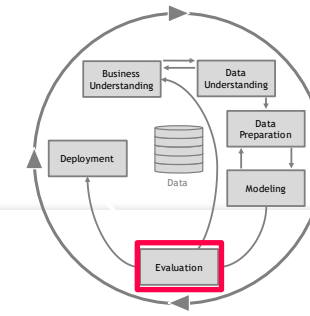


THA



MachineLearningMastery.com, 2017

Evaluation is a crucial step prior to the final deployment of the model



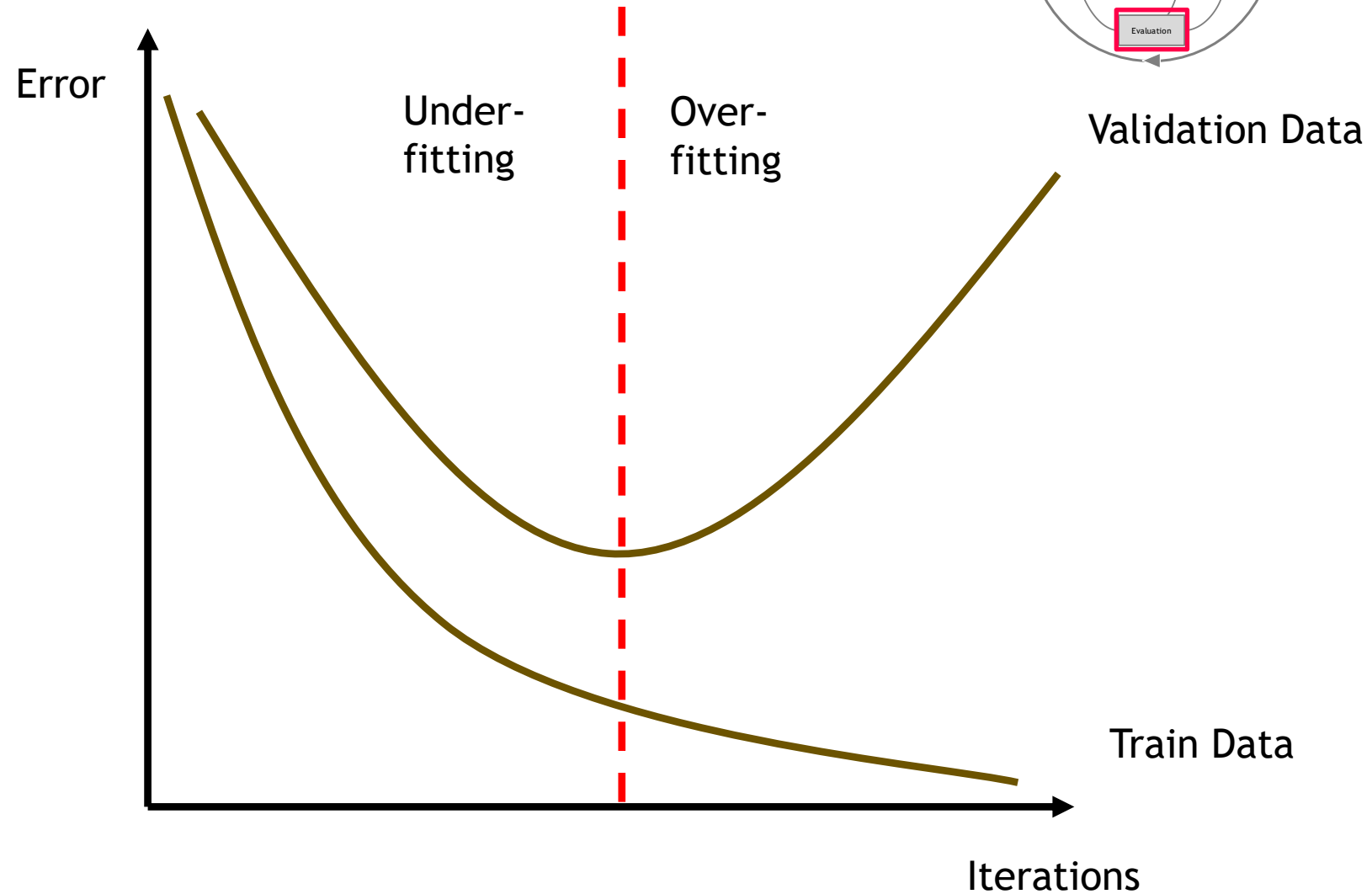
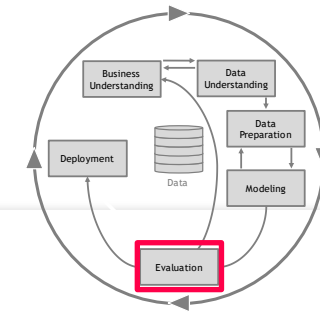
Evaluation of results

**Review of the modeling
process**

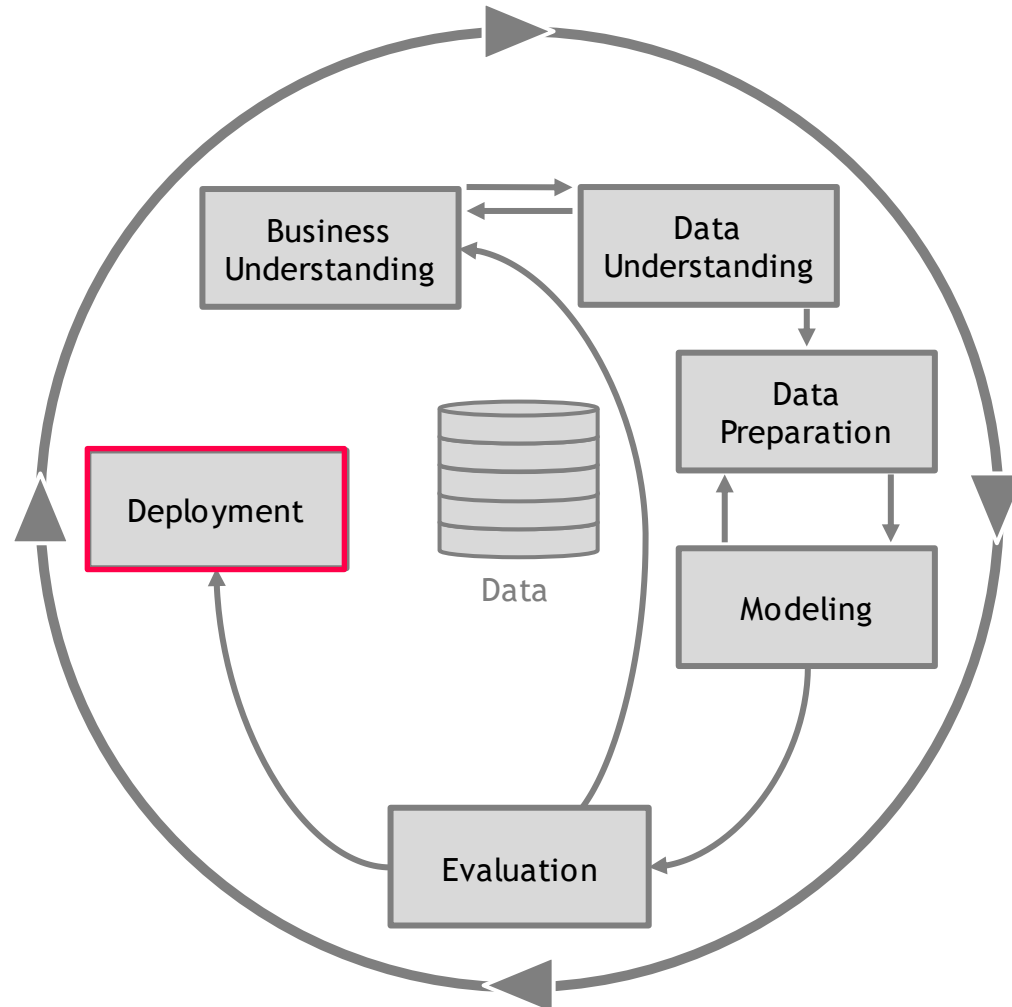
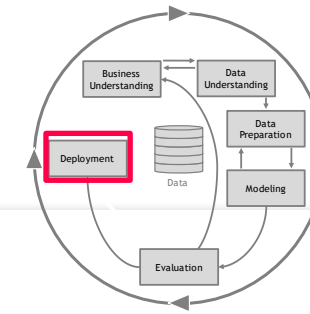
Determination of next steps

[Based on Chapman et al. (2000)]

Tradeoff between generalisation vs. specificity



Finally, deployment requires more than just creating the model



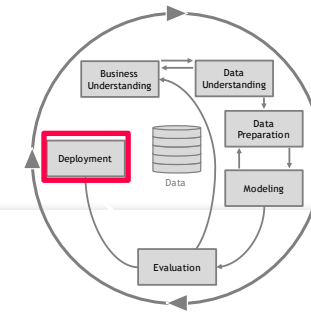
Application of results

Organization of knowledge

Monitoring, maintenance and optimization

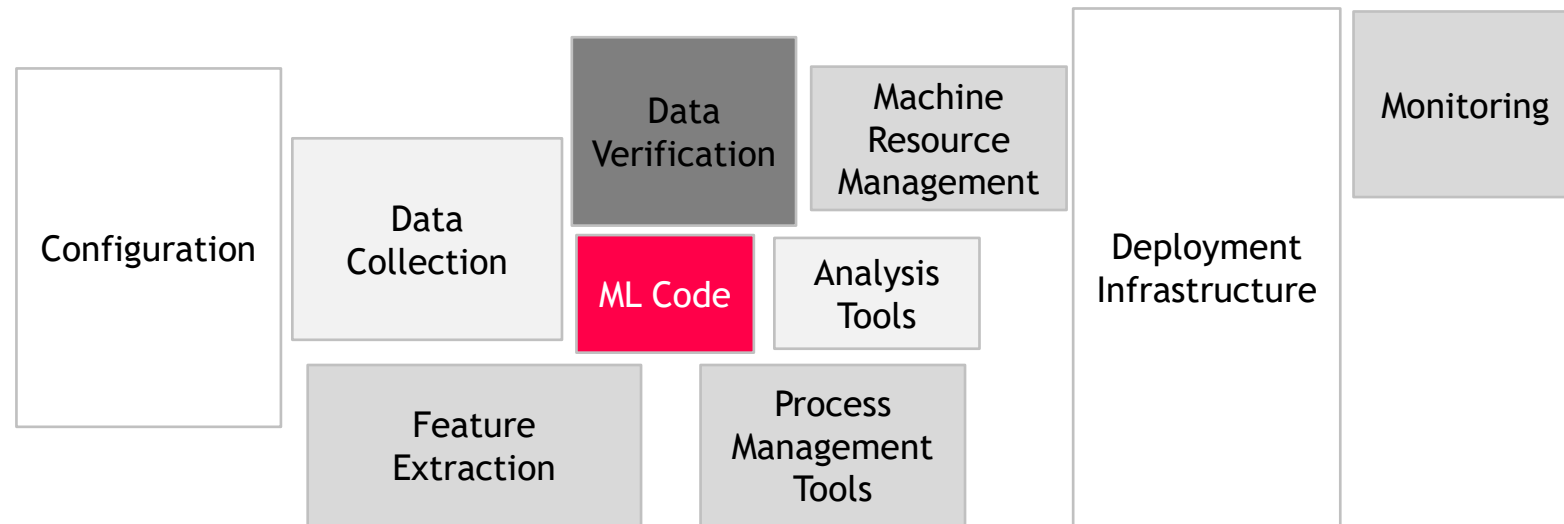
[Based on Chapman et al. (2000)]

Due to the technical complexity, Big Data Analytics systems are often heavily entangled



THA

Main topic in
„Applied AI“
course (IIS4)



In many real-world Machine Learning systems only a small portion of code is direct towards learning and prediction purposes. Due to procedures that are driven by use cases, agility and iteration, systems that incorporate machine learning methods are often subject to technical debt

Contents of today's lecture



What is Data Analytics and how to characterize it?

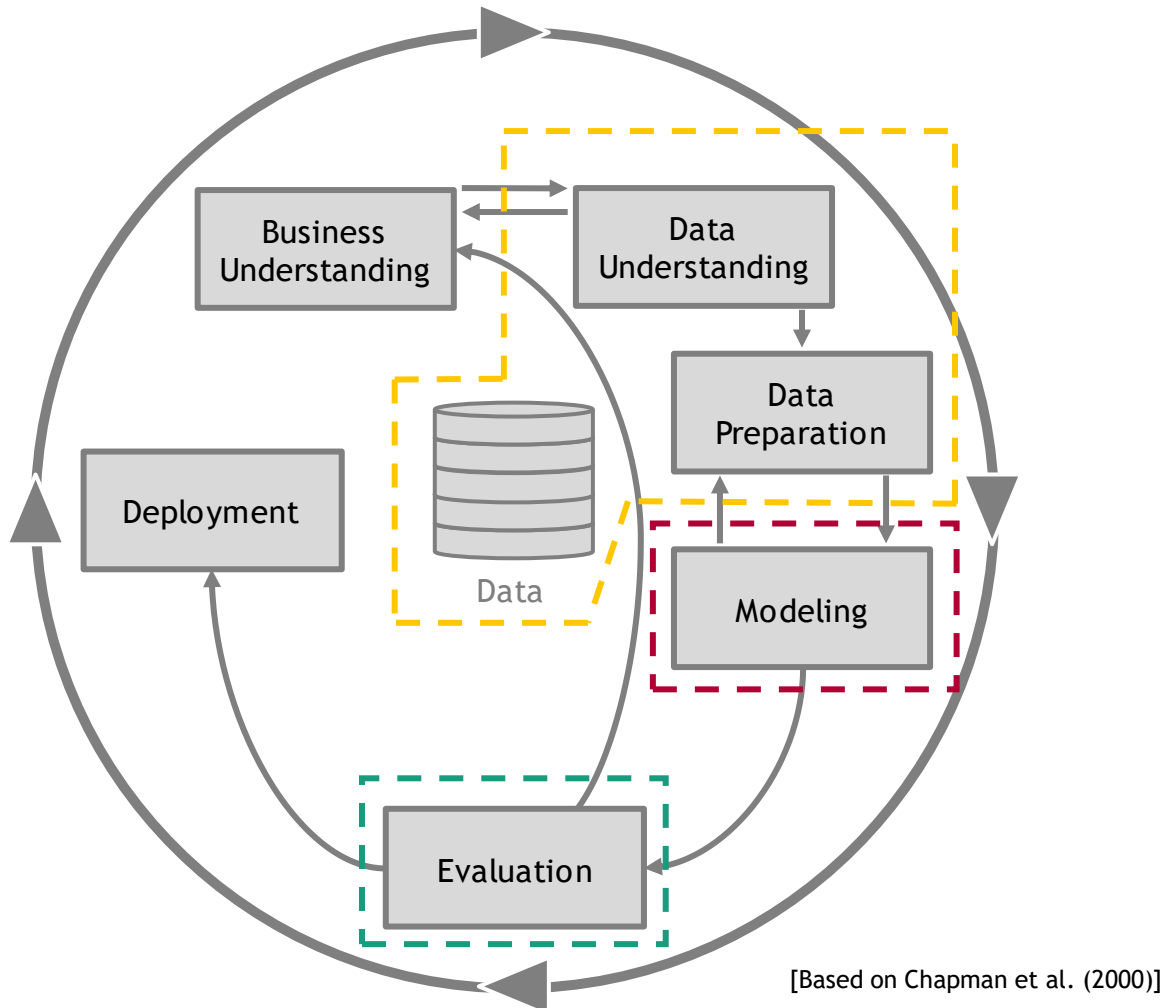


What is Data Analytics and how can we classify AI?



What is CRISP-DM and how do the six steps look like?

Outlook for the course



Slot 1: Introduction to Python



Slot 4: Data Engineering & Management

Slots 5 & 6: Modeling

Slot 7: Evaluation

Slots 8 & 9: Micro Project Phase & Presentations

Slot 10 & 11: Guest Lectures

Slot 12: Advanced Use Cases & Frontier Topics

Contents of today's lecture



What is Data Analytics and how to characterize it?



What is Data Analytics and how can we classify AI?



What is CRISP-DM and how do the six steps look like?

References

- Bellman, R. (1978). *An Introduction to Artificial Intelligence: Can Computers Think?* San Francisco, CA: Boyd & Fraser.
- Chapman, P., J. Clinton, R. Kerber, T. Khabaza, T. Reinartz, C. Shearer and R. Wirth (2000). “CRISP-DM 1.0”. URL: <https://www.the-modeling-agency.com/crisp-dm.pdf> (visited on 11/30/2018).
- Focus (2016). https://www.focus.de/finanzen/news/wirtschaftsticker/unternehmen-microsofts-chatbot-nach-rassistischen-entgleisungen-abgeschaltet_id_5384717.html (Visited 25.05.2020)
- Focus (2018). https://www.focus.de/finanzen/boerse/die-tuecken-der-intelligenz-amazon-schaltet-ki-ab-die-bewerbungen-von-frauen-als-minderwertig-erachtete_id_9741890.html (Visited 25.05.2020)
- Statista (2022). Volume of data. URL: <https://www.statista.com/statistics/871513/worldwide-data-created/>
- Gimpel, H. & Röglin, M. (2017). *Disruptive Technologien - Blockchain, Deep Learning & Co.* Wirtschaftsinformatik & Management
- Handelsblatt (2019). <https://www.handelsblatt.com/technik/sicherheit-im-netz/kuenstliche-intelligenz-was-man-beachten-sollte-wenn-software-die-bewerbung-annimmt/24980886.html?ticket=ST-2541230-bkq7crtZQMyk4Ou2TTLg-ap2> (visited 25.05.2020)
- Haugeland, J. (1987). *Artificial Intelligence: The Very Idea*. 3. print. Cambridge, MA: MIT Press.
- Hofmann, P., Jöhnk J., Protschky, D., Stähle P., Urbach, N., Buck, C. *KI-Anwendungsfälle zielgerichtet identifizieren*. Wirtschaftsinformatik & Management
- Kurzweil, R. (1990). *The Age of Intelligent Machines*. Cambridge, MA: MIT Press.
- MachineLearningMastery.com (2017). URL: <https://machinelearningmastery.com/> (visited 23.05.2020)
- Mauro, A. de, M. Greco and M. Grimaldi (2016). “A formal definition of Big Data based on its essential features.” *Library Review* 65 (3), 122-135.
- Russell, S. J., and Norvig, P. (2016). *Artificial Intelligence: A Modern Approach*, Harlow, UK: Pearson Education Limited.
- Sopra Steria GmbH (2017). POTENZIALANALYSE KÜNSTLICHE INTELLIGENZ, S.46 URL: <https://www.soprasteria.de/docs/librariesprovider33/Studien/potenzialanalyse-k%C3%BCnstliche-intelligenz-2017.pdf> (visited 23.05.2020)
- Sweeney, L. (2003). *That's AI?: a history and critique of the field*. Technical Report CMU-CS-03-106.